Fresh Snapshot Evaluation Before Feedback Collection

10,000 unseen URLs • 5% phishing • frozen models, checkpoints and thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	97.35%	73.93%	72.60%	73.26%	93.11%	78.35%	1.35%	27.40%
HYB-MS +10% GAN	97.75%	78.95%	75.00%	76.92%	92.98%	80.91%	1.05%	25.00%

Outcome

The +10% GAN model improved all thresholded operational metrics and PR-AUC. The strict historical rule was not fully met only because ROC-AUC decreased by 0.13 percentage points.

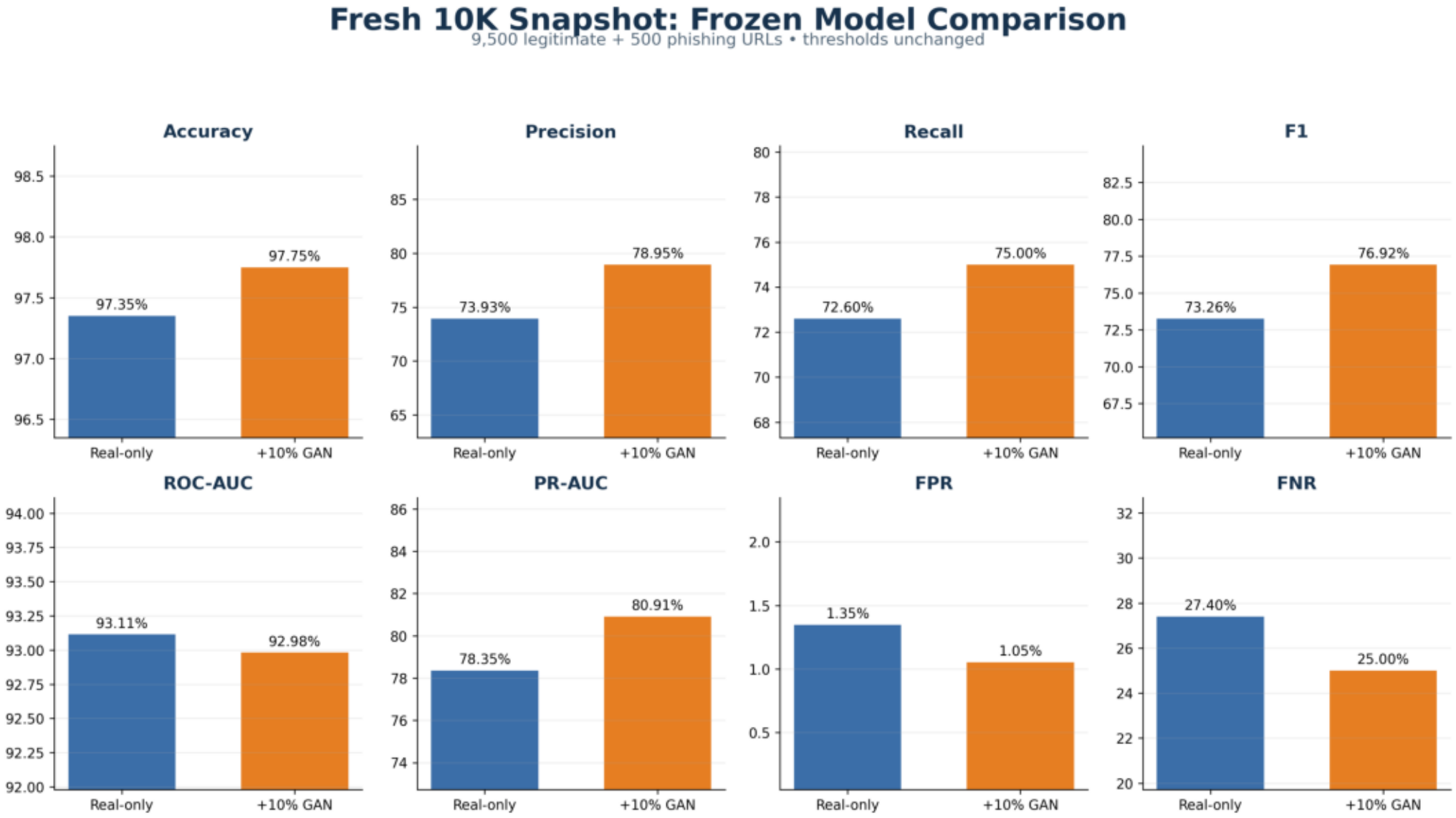
GAN changes: F1 +3.66% • Recall +2.40% • FPR -0.29% • FNR -2.40% • ROC-AUC -0.13%

Deployment gate: neither model reached the registered minimum 80% recall; both stayed below the 2% FPR ceiling.

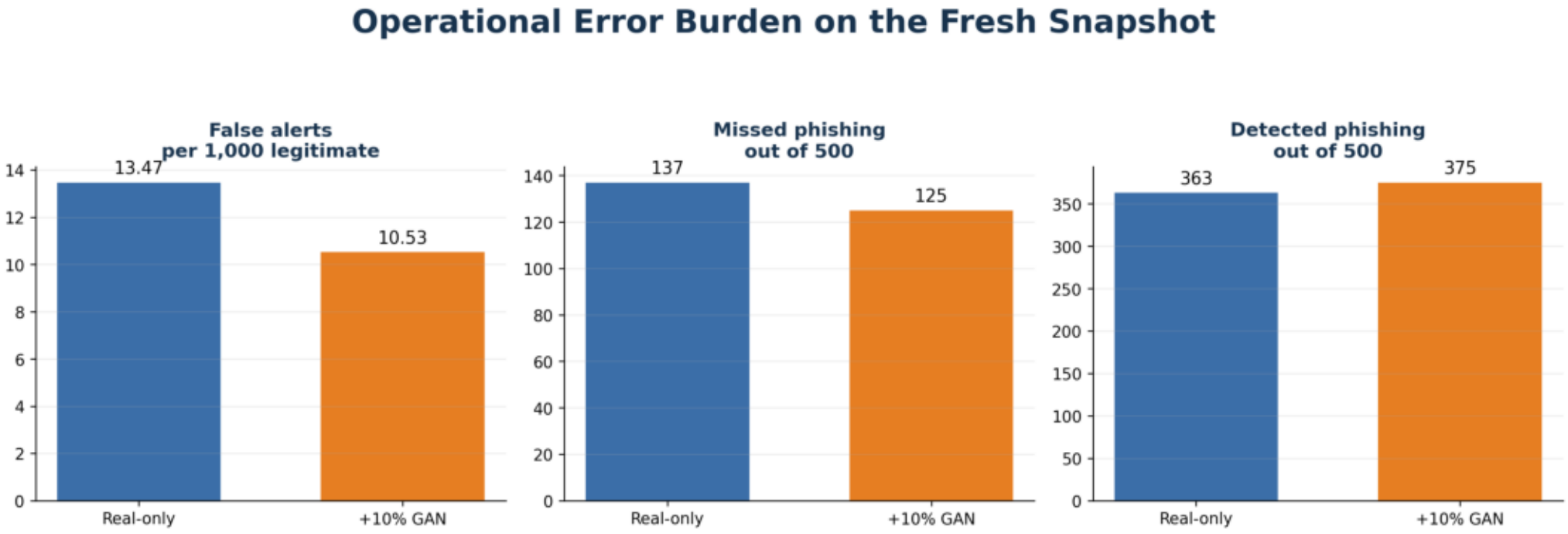
Paired decisions: GAN uniquely corrected 84 rows and uniquely worsened 44 rows; exact McNemar p = 0.0005158.

This cohort was frozen before inference and was not used to alter thresholds or select checkpoints.

All Metrics


Fresh 10K Snapshot: Frozen Model Comparison
9,500 legitimate + 500 phishing URLs • thresholds unchanged
Accuracy
98.5
98.0
97.5
97.0
96.5
97.35%
97.75%
Real-only
+10% GAN
Precision
85
80
75
70
65
73.93%
78.95%
Real-only
+10% GAN
Recall
80
78
76
74
72
70
68
72.60%
75.00%
Real-only
+10% GAN
F1
82.5
80.0
77.5
75.0
72.5
70.0
67.5
73.26%
76.92%
Real-only
+10% GAN
ROC-AUC
94.00
93.75
93.50
93.25
93.00
92.75
92.50
92.25
92.00
93.11%
92.98%
Real-only
+10% GAN
PR-AUC
86
84
82
80
78
76
74
78.35%
80.91%
Real-only
+10% GAN
FPR
2.0
1.5
1.0
0.5
1.35%
1.05%
Real-only
+10% GAN
FNR
32
30
28
26
24
22
20
27.40%
25.00%
Real-only
+10% GAN

Operational Error Burden



Paired Uncertainty and Dataset Integrity

2,000 label-stratified paired bootstrap replicates

Accuracy	**+0.40%**	95% CI +0.19% to +0.62%
Precision	**+5.02%**	95% CI +2.08% to +7.99%
Recall	**+2.40%**	95% CI +0.20% to +4.60%
F1	**+3.66%**	95% CI +1.75% to +5.72%
ROC-AUC	**-0.13%**	95% CI -0.74% to +0.49%
PR-AUC	**+2.55%**	95% CI +1.43% to +3.69%
FPR	**-0.29%**	95% CI -0.49% to -0.11%
FNR	**-2.40%**	95% CI -4.60% to -0.20%

Frozen cohort

9,500 legitimate + 500 phishing • zero URL, hostname and 128-token overlap

SHA-256: 6801eb222c49779fb431f4c45e98e85ac984ae8a6bbd65f1c5c29abdc52cb812